Term 4 answer key

Formulas from memory. Terms 3–4 add the Year 8 families (Pythagoras; trapezium, rhombus and kite, circle area; index laws; percentage change, profit and loss). Answers with π are rounded to 1 decimal place, as each question says; probabilities are fractions in simplest form; every other answer is exact.

WEEK	SESSION	Q1 A Angles & shapes + Pythagoras	Q2 B Measurement	Q3 C Number & percentages + index laws	Q4 D Algebra, ratio & rates	Q5 E Number plane, data & chance
1	1	60°	153.9 m^2	−12	$7y + 21$	6
	2	115°	72 cm^3	$92	50 words per minute	9
	3	42°	50.3 m	90%	25 km/h	4
2	1	110°	65 cm^2	41	$x = 24$	3, 2
	2	110°	30 cm^2	16	$210 and $150	5/6
	3	8 cm	36 cm^2	43 kg	6	9/10
3	1	15 cm	24 cm	16/25	$6x + 10$	1/3
	2	93°	80 cm^2	7^6	07:15	9
	3	50°	50.3 cm^2	5.62	$x = 5$	3
4	1	118°	30 cm^2	50% profit	3 : 2	(4, 5)
	2	55°	5 m	15	24	6
	3	48°	50 cm^2	80%	Bag B ($3 per kg)	4/7
5	1	21 cm	25.1 cm	10	$6a + 6$	7/25
	2	133°	40 cm^2	9%	5 hours	12
	3	13 cm	50 m^2	30 L	$300 and $200	3/8
6	1	48°	78.5 cm^2	51 km	$m = 9$	2, 8
	2	48°	38 cm	15% loss	21	7
	3	65°	7 cm	p^{12}	2 h 35 min	15
7	1	124°	16 cm	5190	$5x + 10$	3/5
	2	123°	96 cm^2	85%	1 : 5	19 cm
	3	16 cm	84 cm^2	−5	$4 per kg	4/5
8	1	20 m	36 cm^3	0.64	$m = 18$	(4, 1)
	2	53°	37.7 m	4	22	1/2
	3	122°	28.3 m^2	$6	2 m/s	14
9	1	140°	96 cm^2	$36	60 and 12	14
	2	110°	70 m	36 m	$12 - 6y$	9
	3	125°	48 cm^2	95%	35 minutes	7/12
10	1	24 cm	66 cm^2	3^6	$x = 8$	3/10
	2	105°	48 cm^2	48	35	(3, 5)
	3	25 cm	14 m^2	0.375	4 : 3	1/2

A MA4-ANG-C-01 complementary and supplementary angles, angles at a point, vertically opposite angles, angles on parallel lines; MA4-GEO-C-01 angle sum of a triangle and a quadrilateral, exterior angle of a triangle

B MA4-LEN-C-01 perimeter, circumference, length units; MA4-ARE-C-01 area of rectangles, squares, triangles, parallelograms and composite shapes, area units; MA4-VOL-C-01 volume of rectangular prisms, volume and capacity units

C MA4-INT-C-01 order of operations with integers; MA4-IND-C-01 powers, square and cube roots; MA4-FRC-C-01 fraction and percentage of a quantity, one quantity as a percentage of another, fractions, decimals and percentages, multiplying and dividing by 10, 100 and 1000*

D MA4-ALG-C-01 substitution, the distributive law; MA4-EQU-C-01 one- and two-step equations; MA4-RAT-C-01 simplifying ratios, sharing in a ratio, rates, best buys, speed; 24-hour time and time differences*

E MA4-LIN-C-01 tables of values, reading points on the number plane; MA4-DAT-C-01 dot plots; MA4-DAT-C-02 mean, median, mode and range; MA4-PRO-C-01 probability, complementary events, experimental probability*

A MA4-PYT-C-01 Pythagoras' theorem (hypotenuse and shorter side; Pythagorean triples)

B MA4-ARE-C-01 area of trapeziums, rhombuses and kites, and circles (this replaces rectangles and unit conversions in Terms 3–4)

C MA4-IND-C-01 index laws (multiply, divide, power of a power) with numerical bases; the questions with pronumeral bases are Stage 5 content (MA5-IND-C-01), included as asked; MA4-FRC-C-01 percentage increase and decrease, percentage profit and loss*

Codes are from the NSW Mathematics K–10 Syllabus (2022). The NESA website could not be reached from the build machine, so **none of these codes was confirmed on the NESA site**; they match the codes used in the Year 9 and Year 10 keys.

* the topic is in this outcome as far as could be checked, but where it sits in the syllabus (or, for 24-hour time, whether it is Stage 3 revision with no Stage 4 code) could not be confirmed.